

Question	Answer	Marks	Guidance
3(a)	sum of potential energy and kinetic energy	B1	Allow 'PE' or ' E_P ' for 'potential energy'. Allow 'KE' or ' E_K ' for 'kinetic energy'.
	(total) energy of random motion of particles	B1	Allow 'distribution' for 'motion'. Allow 'random energy of particles'. Allow 'atoms' or 'molecules' for 'particles'. Do not allow 'system' or 'gas' for 'particles'. Do not allow 'a particle' for 'particles'. Ignore any reference to first law of thermodynamics.
3(b)(i)	$pV = NkT$	C1	Allow $pV = nRT$ and $N = nN_A$.
	$V = (8.7 \times 10^{23} \times 1.38 \times 10^{-23} \times 300) / (2.0 \times 10^5) = 0.018 \text{ m}^3$	A1	Full substitution and answer needed. Unit not needed but any unit given must be correct. Allow '8.31 / 6.02×10^{23} ' for ' 1.38×10^{-23} '.
3(b)(ii)	potential energy of <u>molecules</u> is zero	B1	Allow ' E_P ' or 'PE' for 'potential energy'. Allow 'PE = 0 because there are no intermolecular forces'
	internal energy = $\left(\frac{3}{2}\right) \times NkT$	C1	Allow internal energy = $\left(\frac{3}{2}\right)pV$.
	$= \left(\frac{3}{2}\right) \times 8.7 \times 10^{23} \times 1.38 \times 10^{-23} \times 300$ $= 5400 \text{ J}$	A1	Correct to at least 2 significant figures. (5403). Allow 5.4 kJ if unit is clear.

Question	Answer	Marks	Guidance
3(c)	<u>Any three points from:</u> <u>Up to two points relating to change AB</u> - no change in internal energy - work done = thermal energy transferred - work done on the gas and thermal energy transferred from it <u>Up to two points relating to change BC</u> - internal energy is halved - no work done (on/by gas) - decrease in internal energy = thermal energy removed <u>Up to two points relating to change CA</u> - internal energy is doubled - work done by gas = 1.8 kJ - thermal energy transferred to gas = 4.5 kJ <u>Up to two points relating to the complete cycle ABCA</u> - no overall change in internal energy - overall work is done on gas and thermal energy is removed from it - net work done = net thermal energy transferred	B3	<u>List Rules apply.</u> All numerical values must be given with a unit and be correct to at least 2 significant figures. Ignore points that are simply descriptions of or direct read-offs from the graph. Ignore working space. No ECF. Throughout: Must be clear which aspect of the cycle each point is being made in relation to. Ignore points that are not related to internal energy, thermal energy or work done Ignore symbols unless defined Allow 'heat' for 'thermal energy'. Allow 'internal energy decreases' unless quantified as a factor different than halving. Allow internal energy becomes 2.7 kJ. Allow thermal energy removed = 2.7 kJ. Allow 'internal energy increases' unless quantified as a factor different than two. Allow 'internal energy becomes 5.4 kJ / increases by 2.7 kJ'